

CITYRIDER – INSIGHTS REPORT

Urban Transport Operations Performance Analysis

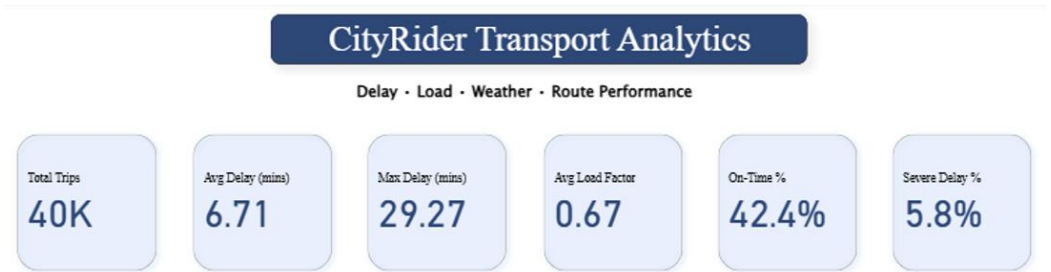
Executive Overview

The CityRider report presents a comprehensive analysis of operational performance across the CityRider public transport network, covering **40,000+ trips**, **50 routes**, and **600+ stops**. The objective of the analysis is to identify delay patterns, demand trends, congestion hotspots, and operational inefficiencies to support data-driven route planning and service optimization.

Key findings indicate that **peak-hour congestion**, **weather disruptions**, and **uneven load distribution** are primary contributors to service delays. The insights generated are consolidated into an interactive Power BI dashboard and a set of actionable operational recommendations.

Key KPIs Overview

KPI	Value
Total Trips Analyzed	40,000+
Average Delay (mins)	6.71
On-Time Reliability	42.4%
Average Load Factor	0.67
Severe Delay Incidents	5.8%



High-Impact Insights

- ✚ **Peak-hour delays** rise sharply during **08:00–10:00** and **17:00–19:00**.
- ✚ **Rainy and stormy weather** significantly increases delay duration.
- ✚ Several routes exhibit **chronic underperformance**, independent of load factor.
- ✚ **Anomaly days** (holidays, festivals, storms) deviate substantially from baseline behaviour.



“Figure 1: Average Delay Trend Over Time”

Route Performance Summary

- ✚ Worst-performing routes combine **long distances** with **dense passenger loads**.
- ✚ “Excellent” routes demonstrate **high reliability**, **stable speeds**, and **low delay variance**.
- ✚ **Load factor** strongly influences delay, particularly along congested commercial corridors.
- ✚ Route performance tiers (Excellent / Moderate / Poor) offer rapid operational classification.

Stop-Level & Passenger Load Findings

- ✚ Certain stops consistently operate **above optimal capacity**, indicating bottlenecks.
- ✚ Heatmaps reveal **high-demand clusters** at commercial and interchange zones.
- ✚ Stop-level congestion shows **direct correlation** with route-level delay spikes.



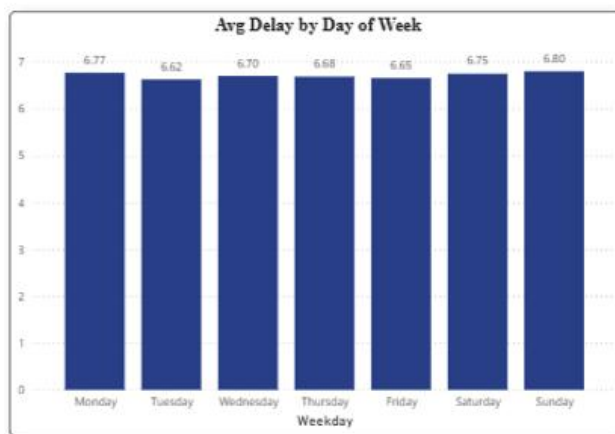
“Figure 2: Route Performance Overview”

Weather & Anomaly Detection

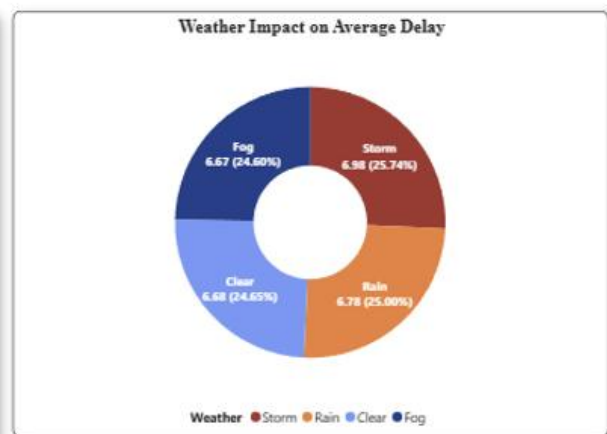
- ✚ Storm conditions cause **large system-wide delay surges**.
- ✚ Anomaly detection identifies **outlier days** where operational patterns diverge sharply.
- ✚ Weather-adjusted forecasting can strengthen **schedule planning and fleet readiness**.

Strategic Recommendations

- ✚ Allocate **1–2 additional buses** on high-delay routes during peak hours.
- ✚ Establish **express routes** for long-distance corridors with overlapping demand.
- ✚ Introduce **buffer scheduling** during adverse-weather days.
- ✚ Reallocate vehicles from **underutilized** to **high-demand** routes.
- ✚ Use clustering insights to prioritize **route redesign** and resource optimization.



Daily delays fluctuate between 6.4–7.2 mins, with visible peaks around mid-month.



Storms & rain increase delays significantly compared to clear days.

Technical Overview

- ✚ Performed advanced data cleaning, timestamp validation, and anomaly treatment.
- ✚ Engineered features including delay metrics, load factor categories, and congestion scores.
- ✚ Conducted EDA with route-level, temporal, and weather-based segmentation.
- ✚ Built an interactive Power BI dashboard with KPIs, route explorer, and drillthrough pages.
- ✚ Developed anomaly detection using statistical thresholds and behavioural deviation.

Conclusion

CityRider demonstrates a complete **end-to-end Data Analytics workflow**—from data acquisition and preprocessing to visualization and operational recommendations. The project reflects real-world analytics capabilities and provides a scalable foundation for future enhancements such as predictive delay modeling, live fleet optimization, and real-time dashboards.